

Olist-ic **Recovery**: ROI Through Intelligence

Maximizing Customer Lifetime Value & Marketing Efficiency via Predictive Targeting in the Brazilian E-Commerce Ecosystem.

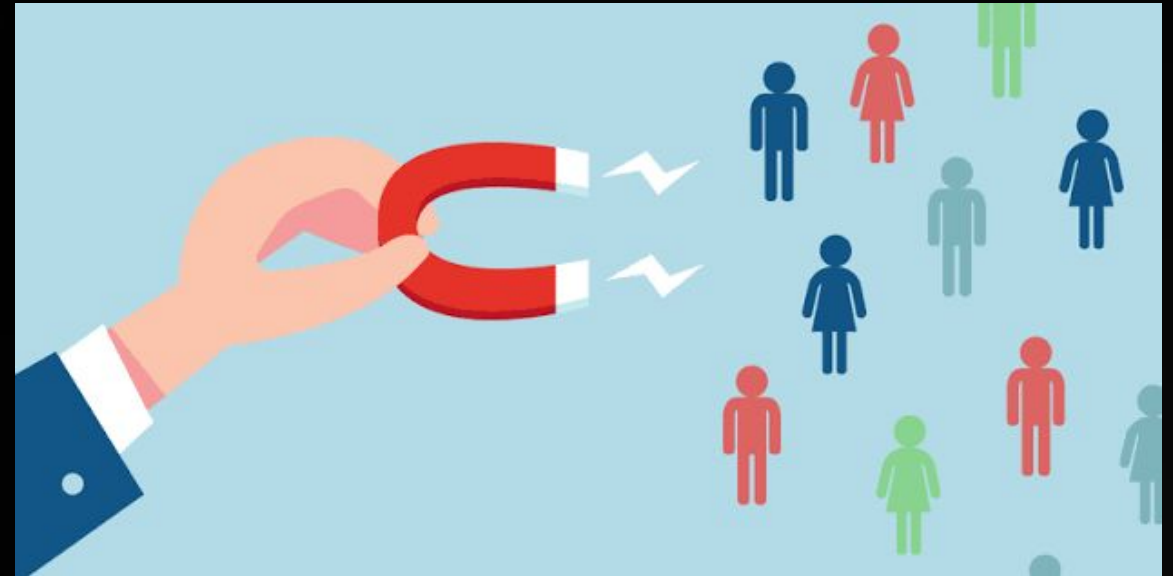
Maika Diaz | PGD in AI and Machine Learning

The Costly Reality of Customer Retention

The Baseline Leak: Within Olist's marketplace, 97% of buyers are one-time purchasers (only ~3.1% repeat within 6 months).

The Marketing Trap: Blanket win-back campaigns are highly inefficient, wasting the bulk of the retention budget on customers who have zero intention of returning.

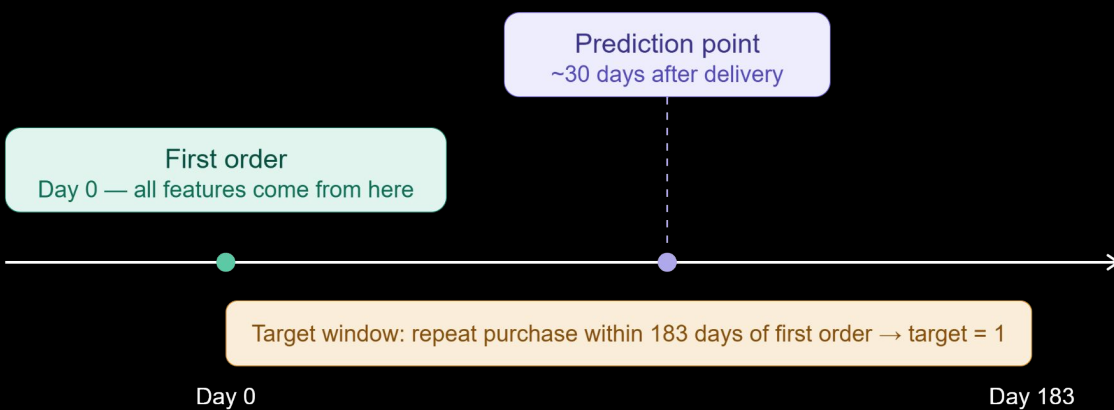
Proposed Strategic Shift: By identifying and isolating the small subset of high-intent repeat buyers, we can concentrate the bulk of our marketing budget directly on them, optimizing customer acquisition costs (CAC) and driving genuine margin growth.



The Data Product Strategy

Predicting and Segmenting Conversion Intent

Model may only use information known by this point — it cannot look forward in time



The Predictive Framework:

- Build an operational model that predicts if a customer will buy again within a 183-day (6-month) window following their initial purchase.

The Operational Safeguard (No Feature Leakage):

- To ensure real-world viability, the model only uses data 30 days after delivery. It cannot look forward in time.

The Retention Advantage:

- Research shows that acquiring a new customer is 5 to 25 times more expensive than retaining an existing one (*Source: Harvard Business Review / Bain & Company*). Focus efforts where the margin is highest.

Model Targeting Performance

Constrained Random Forest

AUC 0.56

Logistic Regression

AUC 0.55

Default Random Forest

AUC 0.53

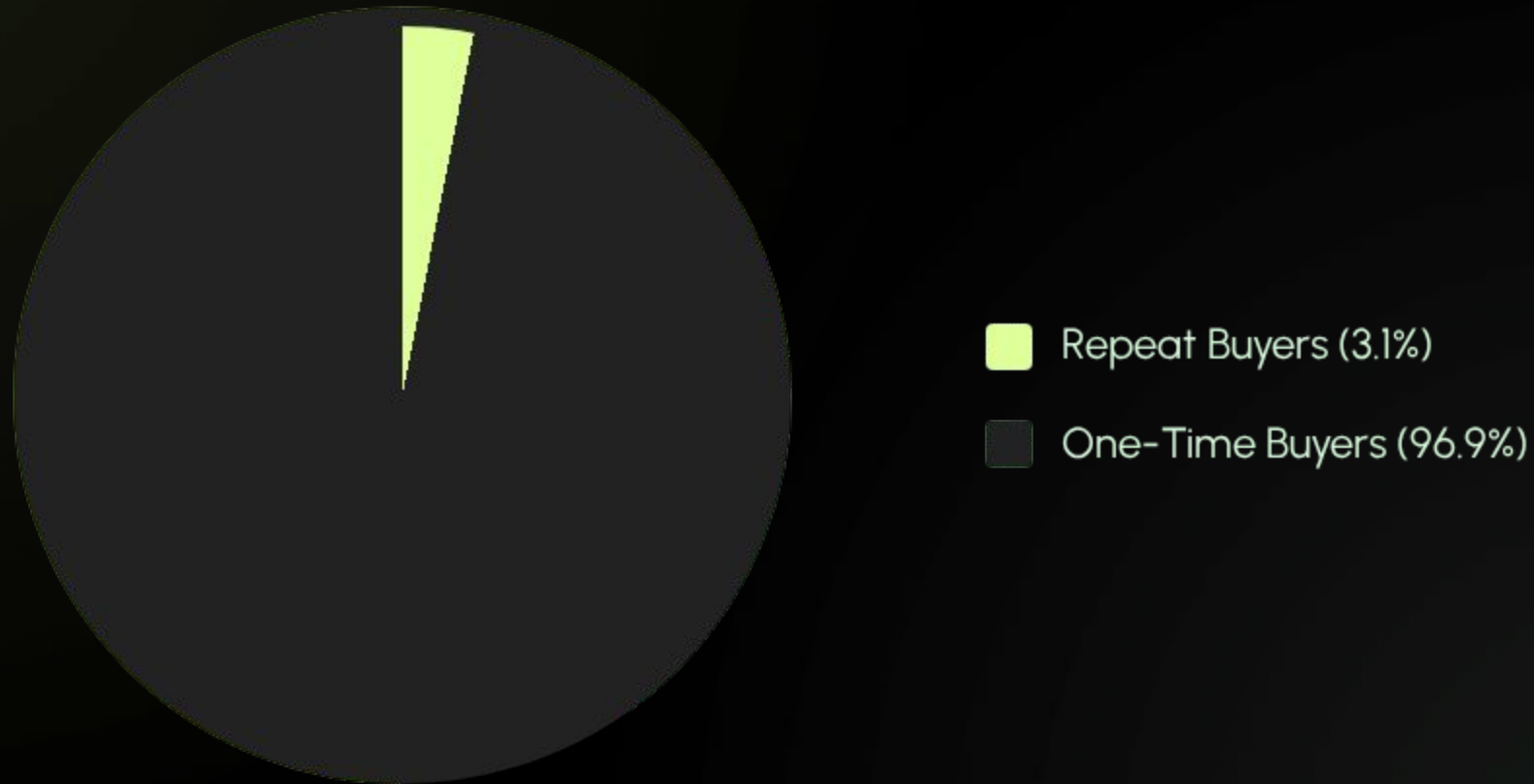
Verdict: The Constrained Random Forest avoids overfitting to noise, providing the best signal for production targeting.

The Honest Surprise: money signal is basket size, not logistics



Despite suffering through terrible logistics, Segment 2 (Burned by Delivery) still converts and repeats at roughly the average marketplace rate, suggesting that delivery friction is unexpectedly not a definitive dealbreaker for customer retention.

Market **Sticky-ness** Reality



Our model targets the critical 3% needle in the haystack, preventing waste on the 97% static user base.

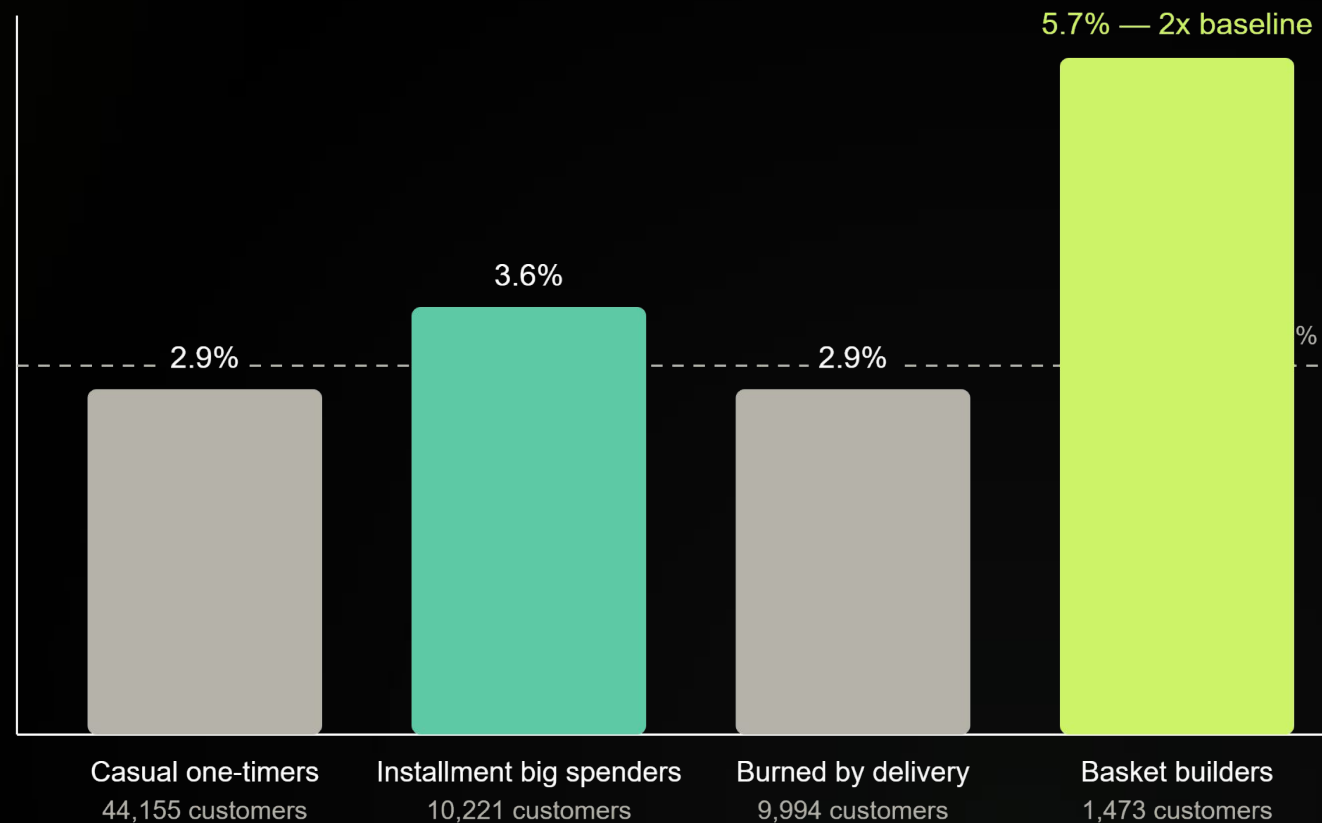
Strategic Segment Profiles

Big Spenders: High ticket values, installment heavy. Loyalty access increases CLV.

Burned by Delivery: 10k users with high friction. Surprisingly, return intent remains at baseline.

Casuals: 44k users. Minimal engagement. Target with high-margin organic reach only.

Repeat-purchase rate by segment (baseline 3.1%)



The Basket Builder Advantage

~2X

Higher Repeat Intent

Unlock Segment Value

While multi-item orders are rare, they are the most actionable insight. "Basket Builders" purchase again at twice the rate of the average customer.

Strategic Action: Automatic VIP onboarding for any customer with a multi-item first cart.

Recommendations



Target the Basket Builder Cohort

Target the top-20% ranked customers with retention spend instead of blanket campaigns



Trigger loyalty treatment on multi-item first carts

Automatic VIP onboarding for any customer with a multi-item first cart.



Invest where it matters

We should not over-invest retention budget in delivery apologies

Responsible AI & Equity



Regional Fairness: I audited model error rates across Brazilian states to ensure no region is systematically ignored.



Dynamic Thresholds: Recommendation to adjust targeting thresholds by state to mitigate socioeconomic bias in poorer logistics corridors.



Continuous Monitoring: Real-time auditing of False Negative Rates in production to maintain equitable service delivery.
